

CHAPTER II

HYBRID BEAMFORMING FOR MM-WAVE SYSTEMS

This chapter presents two solutions to beamforming problem in mmWave systems. 1) A statistical approach for optimal hybrid beamforming for mmWave MIMO system and 2) Deep learning-based hybrid beamforming framework for mmWave MISO system. The chapter provides a basic understanding of different architectures for hybrid beamforming technique. It proceeds with statistical modelling of multi-user MIMO downlink scenario, channel model, and received signal model. A statistical approach for beamforming design is described and proposed an algorithm Extended Simultaneous Orthogonal Matching Pursuit for optimizing hybrid precoder and combiner. Further, a deep learning strategy for beamforming is discussed and presented a DL-based optimal framework for optimal beamforming in a MISO system.

2.1. DEFINITIONS

PRECODING:

In hybrid beamforming, precoding is performed digitally and helps optimize the signal transmission across multiple antenna elements. The goal is to create the desired beamforming vectors in the RF domain while optimizing the data transmission in the baseband.

COMBINING:

beamforming is a technique used in mm-wave (millimeter-wave) communication systems, such as 5G and beyond, to efficiently exploit the high-frequency bands available for wireless communication. In these systems, both analog and digital beamforming methods are combined to achieve a balance between performance and hardware complexity. Specifically, hybrid beamforming combines the benefits of analog beamforming, which is computationally efficient but less flexible, and digital beamforming, which offers flexibility but is often computationally expensive and requires large amounts of power and resources.

RF CHAIN:

The number of RF chains in a hybrid system is much smaller than in a fully digital beamforming system, reducing hardware complexity and power consumption.

CHANNEL STATE INFORMATION:

Accurate channel estimation is crucial for both the digital and analog precoding stages, but at mmWave frequencies, estimating the channel with precision becomes more difficult due to factors like multipath propagation, atmospheric attenuation, and blockage.

THE SPARSITY OF THE CHANNEL:

In mm-wave communication systems, especially with large antenna arrays, the **spatial domain** plays a key role. The channel matrix is often **high-dimensional**, but due to the **sparse nature** of the channel (only a few dominant paths), much of the information in the matrix can be considered as zero or very weak. The sparsity arises from several factors:

RANK OF THE CHANNEL:

The effective rank of the channel is essential for designing hybrid beamforming techniques. If the channel is sparse (e.g., with only a few significant paths), the rank will be low, meaning that only a few spatial dimensions are useful for communication. This can be leveraged by hybrid beamforming systems to reduce the number of RF chains and phase shifters needed, simplifying the hardware and signal processing requirements.

DATA STREAM:

A **data stream** refers to a continuous flow of data that is transmitted or processed in a sequence over time. In the context of communication systems, a data stream typically refers to a sequence of bits, symbols, or packets that are sent from a source to a destination in a continuous, often real-time, manner.

SPECTRAL EFFICIENCY:

Spectral Efficiency (SE) refers to the measure of how efficiently a given communication system utilizes the available **bandwidth** to transmit data. It is typically expressed in **bits per second per Hertz (bps/Hz)**, which represents the data rate per unit of bandwidth. Higher spectral efficiency means more data is transmitted over a given bandwidth, leading to more efficient use of the available

spectrum.

BIT ERROR RATE:

Bit Error Rate (BER) is a key performance metric in communication systems that quantifies the number of received bits that have been incorrectly decoded, relative to the total number of transmitted bits. It provides a measure of the quality of a communication link, where a **lower BER** indicates better system performance with fewer errors in the received data.

MASSIVE MIMO:

Hybrid beamforming is a natural fit for massive MIMO systems in mmWave communications, where the number of antennas can be large but the number of RF chains is kept manageable.

2.2. INTRODUCTION

With the increasing demand for high-speed, low-latency wireless communication, **millimeter-wave (mm-Wave) systems** have emerged as a key technology for **5G** and future **6G** networks. Operating in the frequency range of 30 GHz to 300 GHz, mm-Wave communications offer large bandwidths, which can significantly enhance data rates and system capacity. However, these high-frequency signals face several challenges, including **high propagation loss**, **high sensitivity to environmental factors**, and **limited coverage**. These challenges make the design of efficient transmission systems critical, and **beamforming** plays a crucial role in overcoming them.

To address these challenges and leverage the advantages of mm-Wave communication, **hybrid beamforming** has been proposed as a practical solution, combining both **analog** and **digital** beamforming techniques. This hybrid approach seeks to balance the **performance**, **complexity**, and **energy efficiency** required in mm-Wave systems.

Beamforming is a signal processing technique that utilizes multiple antennas to focus the radio signal in a specific direction, thereby increasing signal strength and reducing interference. In wireless communication systems, beamforming is particularly valuable in achieving high **spectral efficiency** and **improved coverage**, which are essential for mm-Wave communication.

- **Analog Beamforming:** Analog beamforming uses phase shifters or other analog devices to steer the beam at the radio frequency (RF) stage. The goal is to direct the signal towards the intended receiver, enhancing the signal strength at the receiver.
- **Digital Beamforming:** Digital beamforming, on the other hand, processes signals at the baseband stage using **digital signal processing** (DSP). It offers more flexibility and precision since it can adapt the signal to different users, manage interference, and implement more advanced techniques such as **multi-user MIMO** (Multiple Input, Multiple Output).

The primary challenge of mm-Wave communication systems is that they require large **antenna arrays** (with hundreds or even thousands of antennas) to achieve the desired **beamforming gain**. A fully **digital beamforming** approach, where each antenna is connected to its own RF chain, is not feasible due to the **high cost, complexity, and power consumption** of maintaining so many RF chains.

To overcome these limitations, **hybrid beamforming** combines **analog** and **digital** beamforming. This approach typically uses **fewer RF chains** (compared to the number of antennas), with **analog beamforming** performing the directional **multi-user interference** and **spatial multiplexing**.

Key Characteristics of Hybrid Beamforming:

1. **Analog Beamforming Stage:** The analog stage involves RF components, such as **phase shifters**, to steer the beam direction. Since only a few RF chains are used, the system can focus the signal energy in the desired direction, steering and **digital beamforming** handling signal processing for tasks such as improving signal quality and reducing interference.
2. **Digital Beamforming Stage:** The digital stage involves baseband processing to perform tasks like **precoding, multi-user scheduling, and interference management**. Digital processing is used to fine-tune the signal and perform more sophisticated tasks that cannot be easily handled by the analog stage alone.
3. **Reduced RF Chains:** Unlike fully digital systems that require a separate RF chain for each antenna, hybrid beamforming uses a **smaller number of RF chains** (often much fewer than the number of antennas), which results in **lower complexity and energy consumption**.

Benefits of Hybrid Beamforming

1. **Lower Complexity and Cost:** Hybrid beamforming reduces the need for a large number of RF chains, which simplifies the hardware architecture and

lowers costs. This is particularly important in mm-Wave systems, where large antenna arrays are necessary to compensate for the high propagation losses.

2. **Energy Efficiency:** With fewer RF chains and a combination of analog and digital components, hybrid beamforming systems are more power-efficient compared to fully digital beamforming systems, which require power for each RF chain.
3. **Improved Coverage and Signal Quality:** Analog beamforming helps focus energy in the direction of the receiver, overcoming some of the path loss inherent in mm-Wave frequencies. This results in improved coverage, signal strength, and reliability.
4. **Scalability:** Hybrid beamforming allows for the use of large antenna arrays while keeping the number of RF chains manageable, making it a scalable solution for future mm-Wave systems.
5. **Spectral Efficiency:** Digital beamforming enables advanced techniques such as **precoding** and **multi-user MIMO**, which increase spectral efficiency by allowing the transmission of multiple data streams simultaneously to different users.

Precoding and Hybrid Beamforming

In mm-Wave systems, **precoding** is used to optimize the transmitted signal, ensuring that it reaches the desired receivers with minimal interference. Precoding techniques can be performed both in the analog and digital domains:

- **Analog Precoding:** In this stage, phase shifters or other RF components are used to steer the beam in the desired direction. The focus is on directing the energy towards the user or group of users in a way that maximizes received signal strength.
- **Digital Precoding:** Digital signal processing is used to adjust the signals to multiple users, manage interference, and optimize the overall system performance. This stage handles more complex operations that require precise control over the transmitted signals.

Hybrid beamforming combines both these precoding techniques to balance simplicity and flexibility. While analog precoding is effective for simple steering of the beam, digital precoding allows for greater flexibility in managing multiple users and optimizing overall network performance.

Challenges in Hybrid Beamforming for mm-Wave:

1. While hybrid beamforming offers significant advantages, there are still several challenges to **Channel Estimation:** Accurate channel state

information (CSI) is essential for effective beamforming. In mm-Wave systems, this is challenging due to the **sparse nature** of mm-Wave channels and the **high sensitivity** of the system to environmental factors (such as obstacles or mobility).

2. **Beamforming Design:** The design of hybrid beamforming systems requires careful optimization of the analog and digital beamforming stages. This involves trade-offs between system performance (e.g., throughput, coverage) and hardware complexity (e.g., number of RF chains, cost, power consumption).
3. **Interference Management:** In multi-user systems, managing interference efficiently while maintaining high data rates is a complex task. This is especially challenging in hybrid beamforming systems with fewer RF chains and limited digital processing.
4. **Hardware Limitations:** The performance of hybrid beamforming is heavily dependent on the quality and capabilities of the hardware, such as the resolution of phase shifters and the number of RF chains.

Despite the fact that several publications on hybrid beamforming optimization [53–55] exist, they all require perfect CSI or use time-consuming iterative approaches (even at the element level). We can deduce that traditional model-based solutions for mmWave massive MIMO fail to exploit the channel's sparsity features [56]. Due to the inability of conventional methods to utilise structural traits of the channel, low complexity strategies have been adopted, thereby lowering the quality of hybrid precoding. This situation necessitates a disruptive solution that fixes the problem fundamentally while also introducing new approaches to increase precoding quality. There are no systems in place to intelligently adapt beamforming depending on network data in real time. Traditional beamforming methods [57–60] produce good results, but they are complex and prone to complex iterations. The paradigm has moved from a model-based to a data-driven approach, thanks to the recent success of deep learning (DL). Unlike the mathematical (statistical) models employed in modelbased methods, real-time channel information is used to train the neural network (NN) in the data-driven approach [61]. Although, offline training of NN is straightforward

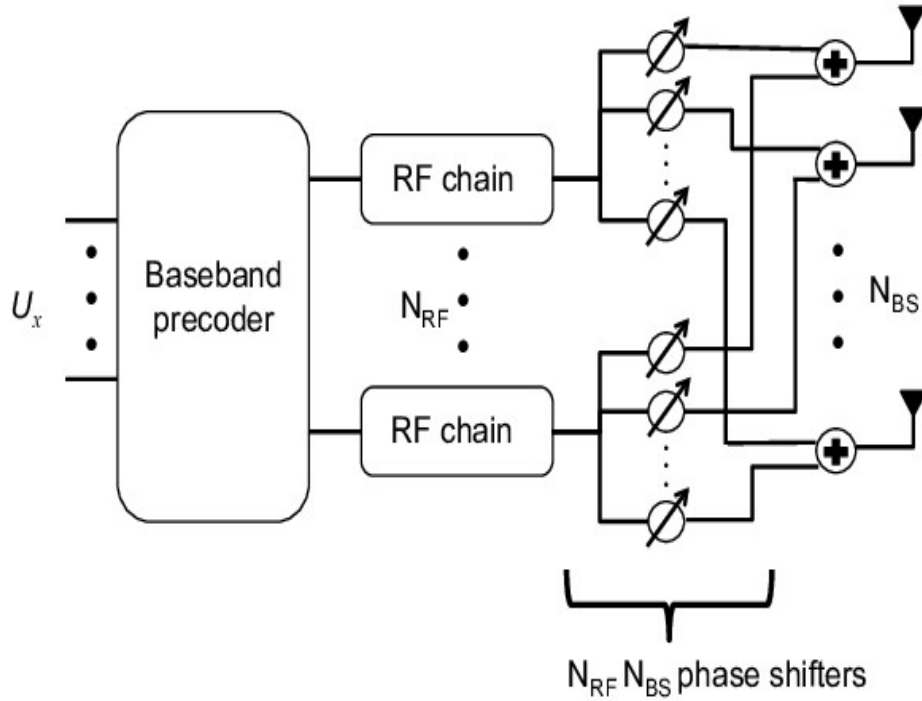


Figure 2.1 Fully Connected

2.3 BACKGROUND

Millimeter-wave (mm-wave) communication is a key enabler for the high data rates and low latency required in next-generation wireless networks (e.g., 5G and beyond). Operating in frequency bands above 24 GHz, mm-wave systems offer significant bandwidth compared to traditional sub-6 GHz systems. However, mm-wave communication faces several challenges, including high propagation loss, signal attenuation due to atmospheric absorption, and limited coverage due to the reduced diffraction and scattering characteristics of mm-waves. One of the most promising techniques to overcome these challenges is **beamforming**.

Beamforming is a signal processing technique used to direct the transmission or

reception of radio waves in specific directions, rather than spreading them uniformly in all directions. In the context of mm-wave systems, beamforming is essential because of the high path loss at these frequencies and the need to achieve high spatial directionality for effective communication.

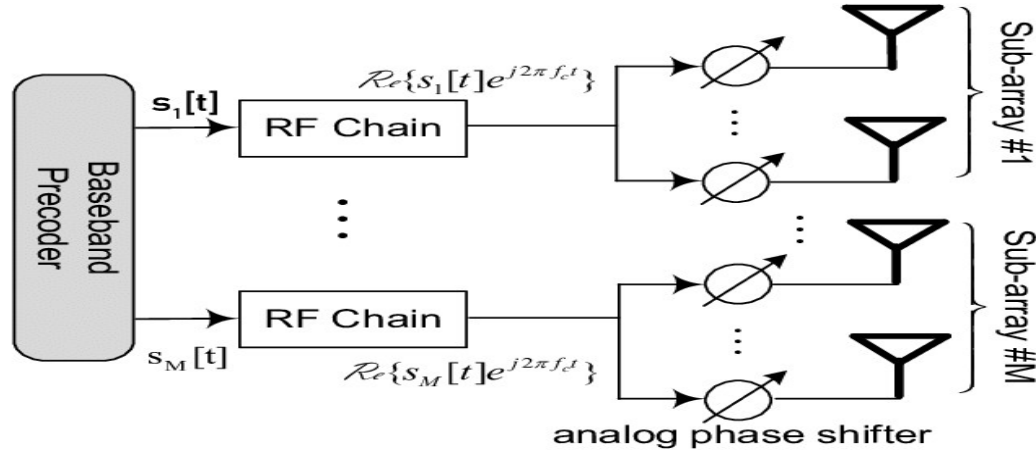


Figure 2.2 Partially Connected

In digital beamforming, the signal is processed at the digital baseband level, and the beamforming weights are applied to the data streams for each antenna element. This approach offers fine-grained control over the signal direction, but it requires a large number of radio-frequency (RF) chains and can be very power- and cost-inefficient, particularly in mm-wave systems with large antenna arrays.

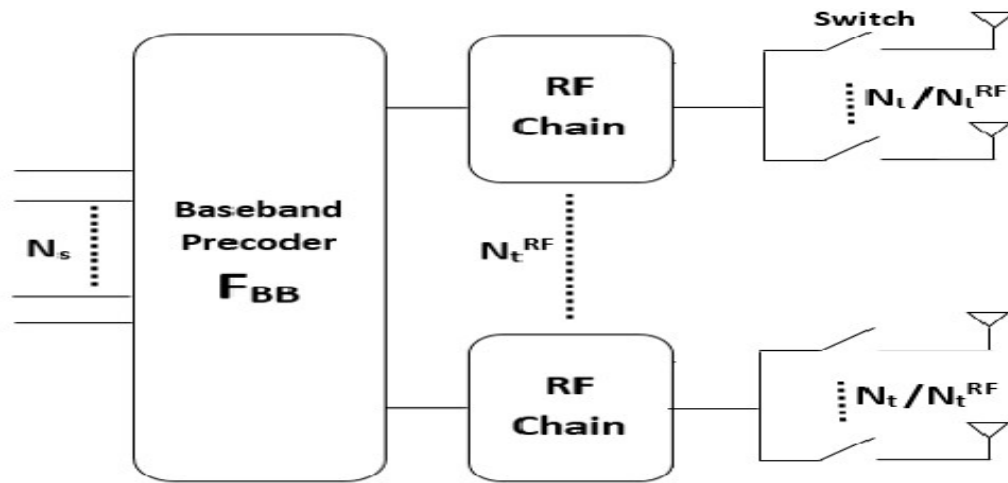


Figure 2.3 Partially Connected Using switch

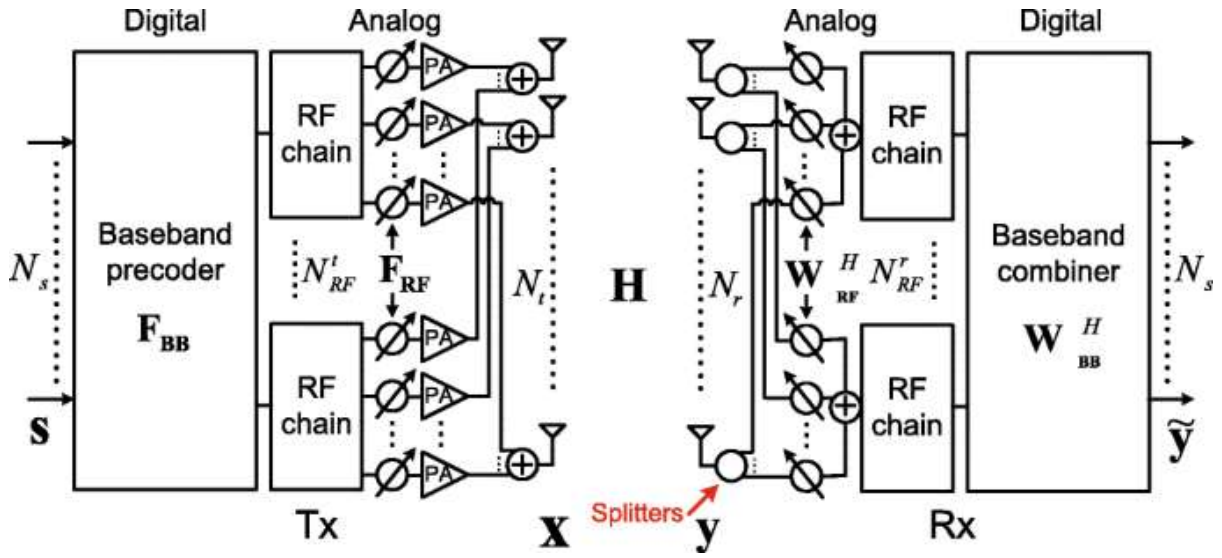


Figure 2.4 Hybrid Beamforming mmWave system

analog beamforming, the beamforming is performed in the analog domain using phase shifters and other RF components. This approach requires fewer RF chains, but it limits the flexibility of the system, as the beam direction is determined by fixed or coarse adjustments to the phase, and multiple data streams cannot be processed independently. Hybrid beamforming is a combination of both digital and analog beamforming. It leverages the strengths of both approaches, with digital beamforming handling the baseband processing and analog beamforming directing the beams in the RF domain. This approach strikes a balance between performance and complexity, offering a feasible solution for mm-wave systems. In mm-wave communication, large antenna arrays are typically used to compensate for the high path loss and to achieve high beamforming gain. However, providing a full digital beamforming solution for each antenna element is expensive and energy-inefficient. Hybrid beamforming uses fewer RF chains, which reduces hardware complexity, cost, and power consumption. By combining the flexibility of digital beamforming with the efficiency of analog beamforming, hybrid beamforming enables high-performance beam steering with a manageable number of RF chains. It allows multiple data streams to be transmitted using fewer RF chains than digital beamforming, while still exploiting the benefits of spatial multiplexing. As mm-wave systems deploy larger antenna arrays (e.g., in massive MIMO configurations), hybrid beamforming offers a scalable solution. It supports large-scale systems without the need for a prohibitively large number of RF chains.

2.3.1. Literature Review on Hybrid Beamforming

Hybrid beamforming has emerged as a crucial technique for overcoming the challenges of millimeter-wave (mm-wave) communications, particularly in the context of large-scale antenna arrays and the need for high data rates in next-generation wireless networks such as 5G and beyond. By combining the benefits of both analog and digital beamforming, hybrid beamforming offers a trade-off between performance, complexity, and hardware efficiency, making it an attractive solution for mm-wave systems. This literature review summarizes key research developments and findings in the area of hybrid beamforming, focusing on its applications, design principles, algorithms, and challenges.

1. Introduction to Hybrid Beamforming

Hybrid beamforming, as defined in most works, involves the combination of both analog and digital beamforming stages to optimize communication in mm-wave and massive MIMO systems. It enables the use of large antenna arrays while keeping the hardware complexity and energy consumption at a practical level. The basic architecture consists of an analog beamforming stage that handles phase control and beam steering in the RF domain, and a digital beamforming stage that performs processing at the baseband.

2. Key Concepts and Motivations

The motivation for hybrid beamforming comes from the fact that fully digital beamforming requires one RF chain per antenna element, which is impractical for large antenna arrays used in mm-wave systems. Analog beamforming, on the other hand, is more efficient but lacks flexibility. Hybrid beamforming, by using fewer RF chains, reduces the cost, complexity, and power consumption while still achieving the benefits of large antenna arrays.

3. *Millimeter-Wave Communication and Beamforming Needs*

Due to the high path loss, atmospheric absorption, and limited diffraction of mm-wave signals, beamforming is essential to ensure high signal strength and link reliability. In mm-wave systems, beamforming allows for high spatial selectivity, increasing the system's capacity and reducing interference, which is vital for maintaining high data rates and low latency.

4. Types of Hybrid Beamforming

Hybrid beamforming has two major types depending on how the processing is done:

Analog Beamforming

Analog beamforming typically involves phase shifters or beamforming networks that adjust the phases of the signals transmitted by the antennas. While this reduces the need for multiple RF chains, it limits the system's flexibility since analog beamforming typically supports only one data stream at a time.

Digital Beamforming

In digital beamforming, each antenna element is connected to its own RF chain, and the signal processing occurs in the digital domain. This approach allows for multiple independent data streams, offering high flexibility and better control over the beams, but it becomes computationally expensive as the number of antennas increases.

Hybrid Beamforming

Hybrid beamforming combines both approaches. The analog part handles coarse beam steering, while the digital part provides the flexibility for fine control over the signal. This structure reduces the number of RF chains needed and is more energy-efficient than fully digital systems, making it suitable for large-scale MIMO configurations.

5. Design Principles and Algorithms for Hybrid Beamforming

Several design strategies have been proposed in the literature for hybrid beamforming. These strategies typically aim to optimize the beamforming weights, minimize the total power consumption, or maximize the signal-to-noise ratio (SNR).

Analog and Digital Sub-Arrays

Many researchers have explored different ways to partition the antenna array into analog and digital sub-arrays. For instance, some designs treat a small subset of antennas as analog beamformers (using phase shifters), while the remaining antennas are connected to RF chains for digital beamforming. Such designs focus on optimizing the interaction between the analog and digital parts to achieve the best overall performance.

Hybrid Beamforming with Limited RF Chains

Various algorithms have been proposed to handle the issue of limited RF chains in hybrid beamforming systems. These include:

- **Convex optimization:** Some works (e.g., [1]) have formulated hybrid beamforming as a convex optimization problem, focusing on maximizing the received signal power or minimizing the interference while satisfying hardware constraints.
- **Greedy algorithms:** In cases with limited RF chains, greedy algorithms can be used to select which antennas are connected to RF chains for digital beamforming, based on their contribution to the overall beamforming performance.
- **Approximation-based methods:** These methods aim to approximate the optimal beamforming solution by iteratively solving subproblems that handle the analog and digital beamforming parts separately.

Channel Estimation for Hybrid Beamforming

Channel state information (CSI) plays a vital role in the design of hybrid beamforming systems. Accurate CSI estimation methods are essential to ensure that the hybrid beamforming weights are properly computed. Studies (e.g., [2], [3]) have explored techniques such as:

- **Pilot-based estimation:** Using pilot signals to estimate the channel coefficients for both the analog and digital domains.
- **Compressed sensing:** Given the sparse nature of mm-wave channels (e.g., due to limited scatterers), compressed sensing-based methods have been proposed for efficient channel estimation in mm-wave systems.
- **Non-linear estimation techniques:** Machine learning-based methods, including deep learning and reinforcement learning, have also been explored to improve the accuracy and speed of CSI estimation.

Beamforming Codebook Design

For practical implementation, a codebook-based approach is often used to design beamforming vectors for the analog domain. The codebook consists of pre-defined beamforming directions, and the analog part selects the best beamformer from this set. Several methods for designing codebooks that reduce the search complexity and improve the system performance under hybrid beamforming constraints have been proposed (e.g., [4]).

6. Performance Evaluation

The performance of hybrid beamforming systems is evaluated based on several metrics, including:

- **Spectral efficiency:** A key metric for assessing the capacity of the system, considering both the number of RF chains and the beamforming design.
- **Energy efficiency:** Since hybrid beamforming reduces the number of RF chains, it inherently improves energy efficiency compared to fully digital systems.
- **Computational complexity:** The computational complexity of the algorithms for beamforming design and channel estimation can be a major factor in system performance. Hybrid beamforming algorithms are typically designed to strike a balance between complexity and performance.

Studies (e.g., [5], [6]) often simulate the performance of hybrid beamforming in realistic mm-wave propagation environments, comparing it to fully digital and analog beamforming approaches. These simulations show that hybrid beamforming can provide near-optimal performance while significantly reducing the number of RF chains required.

7. Challenges in Hybrid Beamforming

Despite its advantages, hybrid beamforming faces several challenges:

- **Channel Sparsity:** Although mm-wave channels tend to be sparse, accurately estimating the sparse channel coefficients for both analog and digital beamforming remains a challenging task, especially in dynamic environments.
- **Hardware Constraints:** The analog hardware, including phase shifters and combiners, must be designed to operate efficiently at mm-wave frequencies with low power consumption and high precision.
- **Dynamic Environments:** In real-world scenarios, the channel conditions can change rapidly due to mobility, leading to difficulties in maintaining effective beamforming without incurring high overhead for reconfiguration.
- **Interference Management:** Managing interference in dense deployments (e.g., in urban environments) with large antenna arrays is a critical challenge, and hybrid beamforming strategies need to address this issue effectively.

8. Recent Trends and Future Directions

- **Machine Learning and AI:** Recent studies (e.g., [7]) explore the application of machine learning, especially deep learning, in hybrid beamforming for efficient channel estimation, beam selection, and beamforming optimization. These techniques hold promise for improving the performance and adaptability of hybrid beamforming systems.
- **Integrated Sensing and Communication (ISAC):** The convergence of communication and sensing is a new direction in hybrid beamforming research. By utilizing the same infrastructure for both sensing and communication, systems can achieve more efficient beamforming strategies while also enhancing localization and tracking.

2.3.2. Need of Deep Learning in Hybrid Beamforming

The advent of **deep learning (DL)** has significantly impacted various fields, including wireless communications. In particular, deep learning techniques are being explored to optimize and enhance the performance of **hybrid beamforming in mm-wave systems**. Hybrid beamforming combines analog and digital beamforming to achieve an efficient balance between performance and hardware complexity, especially for systems with large antenna arrays. However, traditional optimization and signal processing techniques often face difficulties in handling the complexities and dynamic nature of mm-wave channels, which are prone to rapid changes and highly sparse characteristics.

Deep learning (DL), particularly **neural networks** and **reinforcement learning**, offers the potential to address these challenges by providing more adaptive, data-driven solutions to beamforming design and channel estimation. Below, we explore the various reasons why deep learning is increasingly being considered for hybrid beamforming in mm-wave systems.

1. Challenges in Hybrid Beamforming for mm-Wave Systems

Hybrid beamforming, while an effective technique, comes with several inherent challenges that deep learning can help address:

Channel Estimation

- **Sparse Channels:** Mm-wave communication channels are often sparse, meaning that most of the path gains are concentrated in a few directions. Traditional channel estimation methods (like least squares or compressed sensing) struggle with efficiently estimating the sparse channel coefficients, especially in the presence of noise and limited pilot resources.
- **Dynamic Environments:** In real-world mm-wave systems, channels can vary rapidly due to mobility, blockage, and environmental factors. Existing algorithms require frequent updates, which incur significant overhead in terms of time and computational resources.

Role of DL:

- **Neural Networks (NNs)**, particularly **Convolutional Neural Networks (CNNs)** and **Recurrent Neural Networks (RNNs)**, can be trained to predict sparse mm-wave channels from limited pilot data or from previous channel state information (CSI), significantly improving estimation accuracy and reducing the overhead of frequent re-estimation.
- DL models can learn patterns from historical channel data and adapt quickly to changing environments, making the system more robust to dynamic conditions.

Beamforming Design

- **Optimization Complexity:** The hybrid beamforming problem often involves non-convex optimization, which is computationally expensive. Optimizing the analog and digital beamforming weights requires careful balancing, and traditional algorithms can get stuck in local optima, particularly when dealing with large-scale antenna arrays.
- **Limited RF Chains:** The constraint of using fewer RF chains in hybrid beamforming systems complicates the optimization process. Traditional algorithms may struggle to effectively allocate limited RF chains while ensuring high beamforming gain and low interference.

Role of DL:

- **Deep Reinforcement Learning (DRL)** can be used to solve the beamforming optimization problem by treating it as a sequential decision-making process, where the system learns to select the best beamforming parameters through interactions with the environment. DRL methods, such as **Q-learning** or **Proximal Policy Optimization (PPO)**, enable dynamic

optimization of both analog and digital beamforming weights, adapting to changing channel conditions.

- **End-to-End Learning:** A deep neural network can be trained to learn the mapping between the input features (e.g., CSI or received signal) and the optimal beamforming vectors. Once trained, the network can directly output the optimal beamforming weights, avoiding the need for complex optimization algorithms.

Interference Management

- In hybrid beamforming systems, interference from other users or adjacent beams can degrade system performance. Interference management often requires sophisticated algorithms to coordinate beamforming directions, especially in dense networks with many users.

Role of DL:

- **DL-based Interference Mitigation:** Deep learning techniques can be used for interference prediction and mitigation. Neural networks can be trained to predict interference levels based on the channel environment and other system parameters, allowing the system to adjust beamforming strategies proactively to minimize interference.
- **Multi-User MIMO (MU-MIMO) Systems:** In MU-MIMO systems, DL can be used to jointly optimize the beamforming for multiple users, considering their locations and channel states, rather than optimizing each user's beamforming independently. This improves overall system capacity and reduces interference.

2. Advantages of Deep Learning in Hybrid Beamforming

Data-Driven Adaptation

- **Learning from Data:** Traditional beamforming methods often rely on explicit mathematical models of the wireless channel, which may not always capture the true nature of real-world environments. Deep learning, on the other hand, can learn directly from real data, which means that it can automatically adapt to complex and evolving channel conditions without requiring detailed model assumptions.
- **End-to-End Learning:** In some cases, deep learning models can be trained end-to-end to optimize both channel estimation and beamforming simultaneously. This approach reduces the need for separate steps in the traditional signal processing pipeline and allows the system to optimize the entire process based on real-world data.

Reduction in Computational Complexity

- **Faster Computation:** Once a deep learning model is trained, the time to compute the optimal beamforming weights or channel estimates is dramatically reduced compared to traditional optimization methods. This is particularly beneficial in real-time systems, where low latency is crucial.
- **Efficient Resource Allocation:** DL-based approaches can also help allocate resources, such as RF chains and power, more efficiently. By learning optimal patterns of resource allocation from training data, deep learning models can adaptively allocate resources in a way that maximizes system performance.

Handling Non-Linearities and Complex Scenarios

- **Non-Linear Optimization:** The relationship between the system's inputs (such as the channel state) and outputs (beamforming weights) is often non-linear and difficult to model explicitly. Deep learning excels in capturing such non-linearities, allowing it to effectively model the complex interactions between the analog and digital beamforming stages.
- **Robustness to Uncertainty:** DL models can be trained to be robust to various uncertainties, such as noise, interference, or incomplete CSI. This makes them particularly useful for hybrid beamforming systems, where incomplete or inaccurate channel state information is often the norm.

Scalability for Large-Scale Systems

- As antenna arrays in mm-wave systems grow larger, the complexity of beamforming increases. The sheer number of parameters to be optimized becomes intractable using traditional methods. Deep learning provides a way to handle these large-scale systems by efficiently learning from data rather than explicitly solving large optimization problems.
- **Scalable Neural Network Architectures:** Deep learning models, especially deep reinforcement learning agents, can scale to handle massive antenna arrays and a large number of users, which are typical in 5G and beyond networks.

3. Applications of Deep Learning in Hybrid Beamforming

Channel Estimation and CSI Prediction

- Deep learning can be employed for **channel prediction** using historical data, where the system learns the evolution of the channel over time. This is especially useful in environments where channel conditions change dynamically due to user mobility, environmental factors, or blockage.

- Models such as **CNNs** and **RNNs** have been applied to predict channel states and CSI, improving the accuracy of channel estimation with fewer pilot resources and less overhead.

Beamforming Optimization

- **Deep Reinforcement Learning** has been used to optimize beamforming in a dynamic environment. The system interacts with the wireless channel, learning to adjust beamforming strategies in real-time based on user location, mobility, and interference levels.
- **Supervised Learning**: In some cases, supervised learning methods like **autoencoders** or **generative models** can be used to predict the best beamforming vector based on input features like CSI, reducing the need for real-time optimization.

Multi-User and Massive MIMO Systems

- **DL-based Multi-User Beamforming**: In MU-MIMO systems, DL techniques can optimize beamforming jointly for multiple users, ensuring efficient use of available resources and improving system capacity.
- **Massive MIMO**: Deep learning can efficiently manage the complexity of **massive MIMO systems**, where traditional beamforming methods become computationally prohibitive. Neural networks can learn the optimal strategies for such large systems by analyzing large-scale data from training samples.

4. Challenges and Future Directions

Data Requirements

- Deep learning models require large amounts of training data to generalize well. Collecting this data in real mm-wave environments can be challenging, especially when deploying the system in a new or rapidly changing environment.

Model Interpretability

- Deep learning models, especially deep reinforcement learning agents, often function as "black boxes," making it difficult to understand how and why certain beamforming decisions are made. This lack of interpretability can be a limitation in critical communication systems where transparency is important.

Real-Time Processing

- While deep learning models can reduce computational complexity once trained, the training process itself can be resource-intensive. Additionally, ensuring that deep learning models work efficiently in real-time scenarios (especially with massive MIMO arrays) remains an area of active research.

2.3.3. Literature Review on Deep Learning based Hybrid Beamforming

Hybrid Beamforming Overview

Hybrid beamforming leverages the advantages of both **analog beamforming** (using phase shifters for beam steering) and **digital beamforming** (using digital signal processing). In hybrid systems, the analog part typically handles beamforming with a reduced number of RF chains, while the digital part processes the signals in the baseband, improving flexibility and capacity without requiring one RF chain per antenna.

However, traditional hybrid beamforming suffers from issues like:

- **Non-convex optimization:** The hybrid beamforming problem is typically non-convex and computationally expensive to solve.
- **Channel estimation:** Mm-wave systems often operate in sparse channels, making traditional channel estimation techniques (such as compressed sensing or pilot-based methods) less efficient.
- **Interference management:** In multi-user scenarios or dense deployments, optimizing the beamforming to avoid interference while maintaining high data rates is a significant challenge.

Deep Learning in Hybrid Beamforming

Deep learning can tackle these challenges by leveraging its ability to **learn complex patterns** in data, adapt to dynamic channel conditions, and optimize high-dimensional functions in real-time. Neural networks, especially **Convolutional Neural Networks (CNNs)**, **Recurrent Neural Networks (RNNs)**, and **Deep Reinforcement Learning (DRL)**, have been widely investigated for improving beamforming strategies, channel estimation, and interference management in hybrid systems.

The DL techniques utilised to implement mmWave beamforming are summarised in table.2.2. The work described in [98] offers a framework that integrates massive MIMO into DL to leverage the channel characteristics and sparsity of mmWave MIMO channel. The literature on the channel characteristics that influence the performance of beamforming strategy is summarized in table 2.3. These techniques adopt the strategy of first training the DNN offline with predefined sample data in various channel conditions, the framework seeks to learn the channel's statistics and predict the direction of arrival (DOA). The online learning process can then retrieve relevant output data based on real time input. The work validates the method's robustness for poor CSI, but it doesn't address how to get appropriate beamforming weights. A DNN application for signal detection in a multipath fading channel is proposed in the publication [99]. The DNN model assumes perfect CSI without estimating the channel, and its system performance is compared to the best maximum likelihood detection in the low SNR region. Motivated by these findings, we are focusing on using DL technology to address a complex beamforming design challenge with hardware limits and a CSI that is not perfect. As a result, focus of this research is on developing a system that incorporates DL into hybrid precoding. This chapter presents our contributions towards maximizing system performance in two ways: the statistical approach and the Deep Learning approach. In the statistical method, we achieve the objectives through the following contributions. The transmitter's entire coverage area is divided into angular grids. The transmit array response matrix is calculated by generating a dictionary matrix with the angular grids taken into account. Estimation of a sparse block matrix is used to formulate the precoder optimization issue. The nonzero rows of the sparse block matrix are extracted to find the best baseband precoder. The columns of the array response matrix, which correspond to the nonzero rows of the block spark matrix, make up the RF precoder. 42 The receive array response matrix is computed using the least-squares solution to find the RF combiner. The RF combiner and the channel covariance matrix are processed to create the baseband combiner. The proposed scheme is applied to both uniform linear and planar array antenna systems, and the results are provided for both. Similarly, the contributions towards achieving a robust beamforming solution using the Deep learning method are mentioned below. We can't completely employ typical DL methods because the analog beamformer inherits the constant modulus constraint imposed by analogue phase shifters. As a result, we're using a BlackBox (a multi-layer NN) to replace the analogue beamformer and train it throughout the whole communication channel. We propose a new DL model that uses a two-level design approach to generate optimal beamforming weights that maximize spectral efficiency in both perfect and imperfect CSI circumstances. We discuss the research on the suggested framework that integrates intelligence into hybrid precoding to maximize spectrum efficiency in mmWave MISO systems, as well as the system model for hybrid precoding for mmWave massive MIMO

systems. Deep learning is used instead of statistical methods (which are less robust to flawed CSI) to achieve intelligence. Deep learning deals completely with imperfect CSI and solves a non-convex difficult task. The suggested DL technique uses a two-level design approach to effectively exploit the mmWave channel's sparsity features. A deep neural network also requires extremely few computations once the model has been trained. It excels at exploiting GPU processors to speed up computations and lower loss function as a result of enhanced spectral efficiency.

2.4. Open Research Issues

Spectral efficiency is the art of fitting a large amount of information in given channel bandwidth. Therefore, enhancing the data rate and retaining the spectral consistency efficiency under varying channel conditions is the main requirement of a communication system. Towards this aim, below are the main research Issues. Exploiting the channel characteristic and utilizing them in the right way to maximize the spectral efficiency, demystifying the sparsity property of the channel and computing corresponding block sparse channel matrix is a channel issue. Developing an intelligent system that is robust and attains consistent performance even in imperfect channel conditions that causes uncertainty in the channel properties.

2.5. Scope and Delimitation

This chapter investigates beamforming optimization for improving the spectral efficiency of hybrid beamforming mmWave systems. First, we examine a statistical approach involving the study of signal processing techniques to optimize the procedures in mmWave HBF MIMO systems for multiuser downlink scenarios. Second, we propose an algorithm for the optimal design of precoding and combining that enhances system performance. In the latter case, we study the deep learningbased beamforming framework to achieve robust performance in the mmWave MISO systems. The statistical approach considers 64 antennas as a massive MIMO antenna system in two polarizations: uniform linear and planar fashions. The operating frequency is fixed to 28GHz (as the mmWave band), and a maximum of 6 users (in the multiuser scenario) are considered. The proposed algorithm is limited to enhancing spectral efficiency and improving the BER performance. Furthermore, it assumes perfect CSI is available at the transmitter. Therefore, the investigation of parameters such as overhead and security aspects is

2.4. Multiuser Downlink MIMO System Model

In a HBF structure, the BS adjusts the data streams at baseband using $N_t^{RF} \times N_s$ digital precoder, F_{BB} , where $N_s = Kd$. These modified data streams are converted to RF domain at mmWave frequencies by up conversion process in N_t^{RF} RF chains. Then, the BS uses $N_t \times N_t^{RF}$ analog precoder, F_{RF} with constant modulation constraint $|F_{RF}(i, j)|^2 = 1$, to compute the final transmitted signal. The transmitted signal can be represented as in equation 2.1.

$$X = F_{RF} F_{BB} s = \sum_{l=1}^K F_{RF} F_{BB_l} s_l \quad (2.1)$$

Where $F_{BB} = [F_{BB_1}, \dots, F_{BB_K}]$, and $s \in \mathbb{C}^{N_s \times 1}$ is a column vector with data symbols belonging to all users being served by BS. $s = [s_1^T, \dots, s_K^T]^T$, here s_l is data symbol vector for user l . It is considered that $E\{ss^H\} = I_{N_s}$. Notably, the digital precoder can control the signal amplitude and phase, whereas the analog precoder can only provide the control to modify the signal phase. Thus, elements of F_{RF} are normalized such that $|F_{RF}^{(i,j)}| = \frac{1}{\sqrt{N_t}}$ where $|F_{RF}^{(i,j)}|$ is the (i, j) th entry of F_{RF} pertaining to amplitude.

2.4.1. Millimeter wave channel model

The mmWave channels render limited scattering [100], [101]. Considering these characteristics, the channel model includes L_u independent multipath for each H_k , where $L_u < N_t$ for limited scattering. Then the channel between BS and each UE can be represented as

$$H_k = \sqrt{\frac{N_t N_r^k}{L_u}} \sum_{l=1}^{L_u} \alpha_{k,l} \cdot \mathbf{a}_{UE}(\theta_{k,l}^{UE}, \phi_{k,l}^{UE}) \mathbf{a}_{BS}^H(\theta_{k,l}^{BS}, \phi_{k,l}^{BS}) \quad (2.2)$$

Where $\alpha_{k,l}$ is the complex path coefficient and is considered to

follow the standard, i.i.d, complex Gaussian distribution that produces the Rayleigh component. $\mathbf{a}_{UE}(\theta_{k,l}^{UE}, \phi_{k,l}^{UE})$ and $\mathbf{a}_{BS}^H(\theta_{k,l}^{BS}, \phi_{k,l}^{BS})$ are the steering vector at sender and receiver, respectively, where θ and ϕ are azimuth (elevation) angles of arrival and

departure (AoAs/AoDs) of l^{th} path. For uniform linear arrays (ULAs), only the azimuth AOAs and AoDs are considered, and the array response vector is given by

$$a_{ULA}(\theta) = \frac{1}{\sqrt{N}} [1, e^{jkd \sin(\theta)}, \dots, e^{j(N-1)kd \sin(\theta)}]^T \quad (2.3)$$

Where $k = \frac{2\pi}{\lambda}$ With λ as the wavelength of the carrier signal, and d is the spacing between the antenna elements. For uniform planar array antennas, the array response vector is given by

$$a_{UPA}(\theta, \phi) = \frac{1}{\sqrt{N}} [1, e^{jkd(m \sin(\phi) \sin(\theta) + n \cos(\theta))}, \dots, e^{jkd((W-1) \sin(\phi) \sin(\theta) + (H-1) \cos(\theta))}]^T \quad (2.4)$$

Where $0 \leq m \leq W-1, 0 \leq n \leq H-1$, and $N = WH$. $N = N_t$ for a_{BS} and $N = N_r$ for a_{UE} of H_k .

2.4.2. Received signal model

For k^{th} , the model of received signal can be written as

$$\mathbf{y}_k = \mathbf{H}_k \mathbf{F}_{RF} \mathbf{F}_{BB_k} \mathbf{s}_k + \mathbf{H}_k \sum_{l \neq k} \mathbf{F}_{RF} \mathbf{F}_{BB_k} \mathbf{s}_l + \mathbf{n}_k \quad (2.5)$$

Where $\mathbf{H}_k \in \mathbb{C}^{N_r \times N_t}$ is the channel matrix that contains the complex channel gain for every path between the BS to k^{th} user and $\mathbf{n}_k \sim \mathcal{CN}(0, \sigma^2 \mathbf{I}_{N_r})$ denotes additive white Gaussian noise.

Note that $\mathbf{H}_e = \mathbf{H} \mathbf{F}_{RF}$ is the baseband precoder channel matrix, which can be represented as [61]

$$\mathbf{H}_e = \begin{bmatrix} (f_{RF}^{11}) & (f_{RF}^{12}) & \dots & (f_{RF}^{1N_t^{RF}}) \\ (f_{RF}^{21}) & (f_{RF}^{22}) & \dots & (f_{RF}^{2N_t^{RF}}) \\ \vdots & & & \vdots \\ (f_{RF}^{N_r 1}) & (f_{RF}^{N_r 2}) & & (f_{RF}^{N_r N_t^{RF}}) \end{bmatrix}$$

$$\begin{matrix} \vdots & \diagdown \\ & \dots \end{matrix}$$

Essentially, the entries of $\mathbf{H}_e$ matrix specify the channel gains for every RF chain used in the system [118]. The columns of the RF precoder corresponding to the index

of non-zero columns of $\mathbf{H}_e$ matrix are selected to produce the block sparse matrix. We then compute the digital precoder by applying the SVD to the block sparse matrix [103].

Once the signal is received, the user k would process it via $N_r \times N_r^{RF}$ analog combiner, $\mathbf{W}_{RF}$ designed by phase shifter such that $|\mathbf{W}_{RF}(i, j)|^2 = 1$. The N_r^{RF} RF chains help in down converting the combiner output to baseband. Further, the digital combiner $\mathbf{W}_{BB_k} \in \mathbb{C}^{N_r^{RF} \times d}$ process the signal as given in equation 2.6.

$$\widetilde{\mathbf{y}}_k = \underbrace{\mathbf{W}_k^H \mathbf{H}_k \mathbf{F}_k \mathbf{s}_k}_{\text{desired signal}} + \underbrace{\mathbf{W}_k^H \mathbf{H}_k \sum_{l \neq k} \mathbf{F}_l \mathbf{s}_l}_{\text{effective interference}} + \underbrace{\mathbf{W}_k^H \mathbf{n}_k}_{\text{effective noise}} \quad (2.6)$$

Where $\mathbf{F}_k = \mathbf{F}_{RF} \mathbf{F}_{BB_k}$ and $\mathbf{W}_k = \mathbf{W}_{RF_k} \mathbf{W}_{BB_k}$ are the overall hybrid precoder and combiner for the k^{th} user, respectively.

The constant modulus constraint of the analog precoder has not allowed proposing any general solution to maximize the data rate in 2.7. [104], [105].

$$R = \log_2 \left(\left| \mathbf{I}_{N_r} + \frac{1}{\sigma^2} (\mathbf{W}_{BB}^H \mathbf{W}_{RF}^H \mathbf{W}_{RF} \mathbf{W}_{BB})^{-1} \mathbf{W}_{BB}^H \mathbf{W}_{RF}^H \mathbf{H} \mathbf{F}_{RF} \mathbf{F}_{BB} \mathbf{F}_{BB}^H \mathbf{F}_{RF}^H \mathbf{H}^H \mathbf{W}_{RF} \mathbf{W}_{BB} \right| \right) \quad (2.7)$$

Therefore, applying the zero-forcing precoding on the digital precoder has been recognized as an alternative remedy for managing the inter-user interference, followed by the analog precoder for phase control. The total sum rate of the HBF system for user k when assumed the Gaussian signalling and represented as in equation 2.8.

$$R_k = \log_2 \left| \mathbf{I}_{N_r} + \mathbf{W}_k \mathbf{C}_k^{-1} \mathbf{W}_k^H \mathbf{H}_k \mathbf{F}_k \mathbf{F}_k^H \mathbf{H}_k^H \right| \quad (2.8)$$

$$\mathbf{C}_k = \mathbf{W}_k^H \mathbf{H}_k \left(\sum_{l \neq k} \mathbf{F}_l \mathbf{F}_l^H \right) \mathbf{H}_k^H \mathbf{W}_k + \sigma^2 \mathbf{W}_k^H \mathbf{W}_k$$

Here, $\mathbf{C}_k$ is the covariance of interference and noise at user k . Further, the capacity of system can be represented as equation 2.9. [106].

$$C = \log_2 \det \left(\mathbf{I}_{N_s} + \frac{\rho}{N_s} \mathbf{H}_e^H \mathbf{H}_e \right) \quad (2.9)$$

Here, $\mathbf{I}_{N_s}$ is the $N_s \times N_s$ identity matrix and ρ denote the average SNR at the receiver

2.7 Problem Formulation

This chapter desires to renew the solutions to maximize the system spectral efficiency by optimally designing the hybrid precoder and combiner in HBF mmWave system. Therefore, we formulate the sum-rate maximization problem under power constraint as;

$$\underset{\mathbf{F}_{RF}, \mathbf{F}_{BB}, \mathbf{W}_{RF_k}, \mathbf{W}_{BB_k}}{\text{maximize}} \quad \sum_{k=1}^K P_k R_k \quad (2.10)$$

$$\text{subjected to} \quad \text{Tr}(\mathbf{F}_{RF}, \mathbf{F}_{BB}, \mathbf{F}_{BB}^H \mathbf{F}_{RF}^H) \leq P,$$

$$|\mathbf{F}_{RF}(i, j)|^2 = 1, \forall i, j$$

$$|\mathbf{W}_{RF_k}(i, j)|^2 = 1, \forall i, j$$

The BS is constrained with a total power budget of P , the weight p_k designates the power allocation to user k ; i.e., the larger $\frac{p_k}{\sum_{l=1}^K p_l}$ implies greater power for user k .

The traditional MIMO channel can be decomposed as $\mathbf{H} = \mathbf{U} \Sigma \mathbf{V}^H$, where the matrices $\mathbf{U}$ and $\mathbf{V}$ are semi-Unitary such that $\mathbf{U}^H \mathbf{U} = \mathbf{I}$ and $\mathbf{V}^H \mathbf{V} = \mathbf{I}$ and Σ is the diagonal matrix of non-negative values. For the classical MIMO, we can set the baseband precoder equal to $\mathbf{V}$ and RF precoder to $\mathbf{I}$ or they can even be interchanged. But for mmWave

massive MIMO, the ideal baseband/RF precoder must be set to $\mathbf{A}_t \mathbf{V}$. The transmit array response vectors in transmit array matrix $\mathbf{A}_t$ form a basis for the column space of $\mathbf{H}^H$. The matrix $\mathbf{A}_t$ need to be estimated either by constructing a dictionary matrix or by applying the principle of scattering. Therefore, the best precoder approximation problem can be formulated as $\max_{\mathbf{F}_{BB}} \|\mathbf{V} - \mathbf{A}_t \widetilde{\mathbf{F}}_{BB}\|_F^2$. At the

receiver, the received vector $\mathbf{Y}$ is acted upon by combining weights. The combiner weights must be computed to reduce the receiver's MMSE. The HBF system aims to deliver greater spectral efficiency, the precoding and combining weights decide the HBF system's performance.

Therefore, the optimization problem is formulated so to compute these weights optimally. As the RF beamforming and combining coefficients are used only to control the direction of beams, there are few additional constraints in the optimization problem to compute these weights. Theoretically, the combination of $\mathbf{F}_{\text{BB}} \times \mathbf{F}_{\text{RF}}$ and $\mathbf{W}_{\text{BB}} \times \mathbf{W}_{\text{RF}}$; that is produced by the HBF method are expected to be near estimates of $\mathbf{F}$ and $\mathbf{W}$, which are computed without any constraints.

2.8. Hybrid Beamforming Solutions

2.8.1. Hybrid Beamforming Design: Statistical Method

This section discusses an effective mechanism that considers the beam grids for computing the array response matrices. Subsequently, we propose an extended simultaneous orthogonal matching pursuit algorithm (ESOMP). The transmit array response matrix can be constructed by considering the angular grid ϕ_T of size G , with $\theta \in \phi_T$, $1 \leq i \leq G$ and $G \geq N_t$. The dictionary matrix $\mathbf{A}$ is constructed as in equation 2.11.

$$\mathbf{A} = [\mathbf{a}(\theta_1) \mathbf{a}(\theta_2) \mathbf{a}(\theta_3) \dots \mathbf{a}(\theta_G)] \text{ and } \mathbf{a}(\theta_1) = \begin{bmatrix} 1 \\ e^{-j\frac{2\pi}{\lambda}d\cos\theta_1} \\ \vdots \\ e^{-j\frac{2\pi}{\lambda}(N_t-1)d\cos\theta_1} \end{bmatrix} \quad (2.11)$$

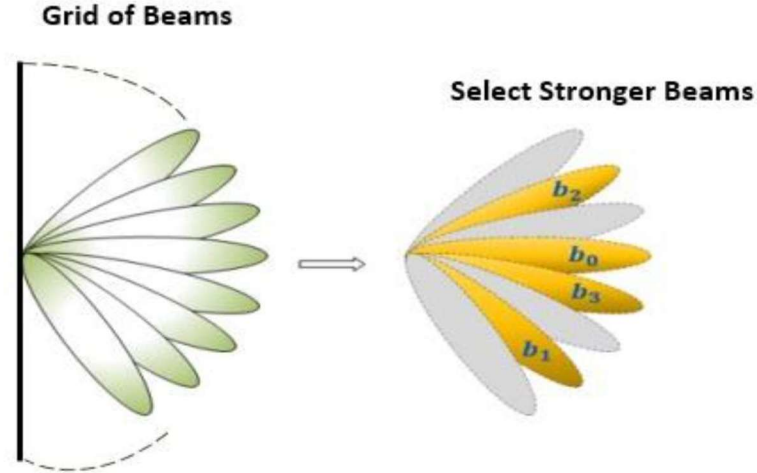


Figure 2.5 Angular Grids

Figure 2.5 illustrates the formation of angular grids across the coverage area of transmission. The BS system chooses only those grids corresponding to the desired directions where the users are located. The base station transmits the signals only in such specific directions. The accurate selection of the beams is possible through the best approximation design of the precoder.

As mentioned earlier, the optimal precoder design can be formulated by developing a block sparse matrix. The non-zero rows of this matrix are extracted to determine the baseband precoder. Further, the RF precoder is developed by extracting those columns from the array response matrix which corresponds to the non-zero rows of the block sparse matrix. Similarly, at the receiver, the analog combiner is determined with the aid of the least square solution on the array response matrix of receiver. Further, the analog combiner matrix is processed along with the channel covariance matrix to design digital combiner.

The proposed scheme is employed for uniform linear and planar array antenna systems, and the performance is presented for both antenna systems. The flow of the research method is as depicted in the flow chart of figure.2.6. The entire process of signal processing to design optimal precoding and combining is sequenced in the algorithms given below.

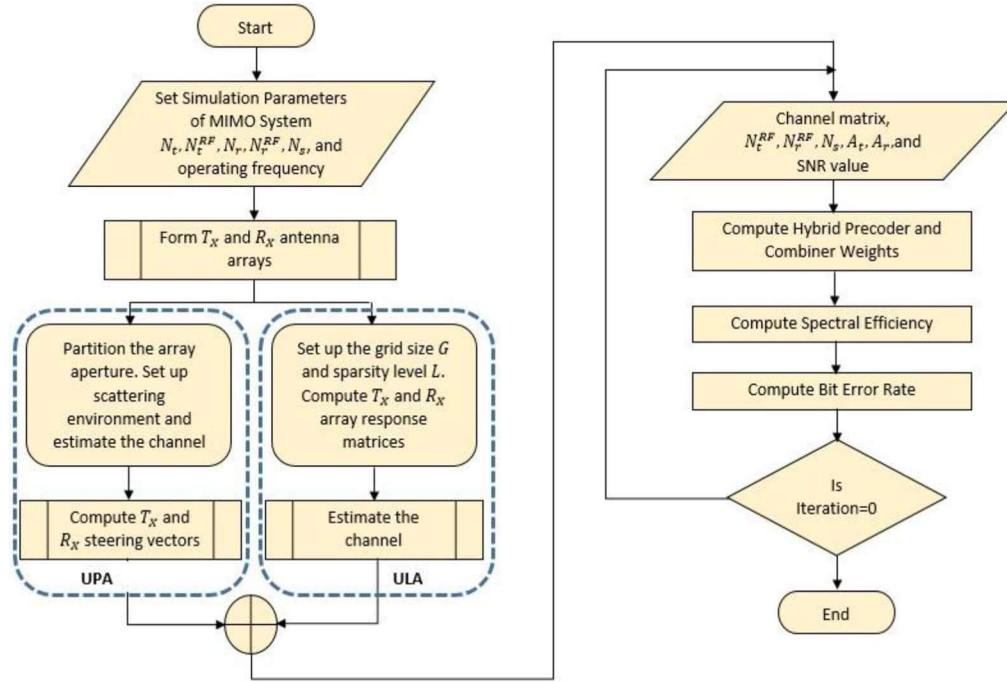


Figure 2.6 Research Flow

Extended SOMP Algorithm: Precoder Optimization

$\mathbf{F}_{R^F} = \mathbf{I}_L$,
 $[\mathbf{U}, \mathbf{S}, \mathbf{V}] = \text{syd}(\mathbf{H})$, $\mathbf{F}_{up} = \mathbf{V}(:, 1:N_s)$
 $\mathbf{F}_{res}^{(0)} = \mathbf{F}_{opt}$
 for $1 \leq k \leq N_{R^F}$
 $\mathbf{A}_H = \mathbf{I}_{N_t} \mathbf{F}_{res}^{(k-1)}$
 $i(k) = \arg \max_i |\mathbf{J}^H \mathbf{I}_i|$
 $\mathbf{F}_{R^F} = \mathbf{A}_t[:, i(k)]$
 $\mathbf{F}_{BB} = \left((\mathbf{F}_{R^F})^H \mathbf{F}_{R^F} \right)^H (\mathbf{F}_{R^F})^H \mathbf{F}_{pt}$
 $\mathbf{F}_{res} = \frac{\mathbf{F}_{gt} - \mathbf{F}_{R^F} \mathbf{F}_{BB}^{(k)}}{\mathbf{F}_{ypt} - \mathbf{F}_{R^F}^{(k)} \mathbf{F}_{BB}^{(k)H} \mathbf{F}_{pt}}$
 end
 $\mathbf{F}_{BB} = \frac{\mathbf{F}_{\emptyset B}}{\mathbf{F}_{R^F} \mathbf{F}_{\emptyset B} \mathbf{F}_{pt}}$
 $\mathbf{F}_{\emptyset B} = \mathbf{F}_{BB}, \mathbf{F}_{R^F} = \mathbf{F}_{R^F}$

Extended SOMP Algorithm: Combiner Optimization

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 $\mathbf{W}_{\text{RF}}^{(0)} = [\ ]$ ,
 $\mathbf{W}_{\text{res}}^{(0)} = \mathbf{W}_{\text{MMSE}}$ 
 $\mathbf{R}_{\text{SS}} = \mathbf{I}_{N_s}$ ,  $\mathbf{R}_{\text{YY}} = \mathbf{H}\mathbf{F}_{\text{RF}}\mathbf{F}_{\text{BB}}\mathbf{R}_{\text{SS}}\mathbf{F}_{\text{BB}}^T\mathbf{H}^T + \text{SNR}^2\mathbf{I}_{N_r}$ 
for  $1 \leq k \leq N_{\text{r}}^{\text{RF}}$ 
     $\boldsymbol{\psi} = \mathbf{A}_{\text{r}}^H \mathbf{R}_{\text{YY}} \mathbf{W}_{\text{res}}^{(k-1)}$ 
     $i(k) = \text{arg max}[\boldsymbol{\psi} \boldsymbol{\psi}^H]_{l,l}$ 
     $\mathbf{W}_{\text{RF}}^{(k)} = \mathbf{A}_{\text{r}}[:, i(k)]$ 
     $\mathbf{W}_{\text{BB}}^{(k)} = \left( \left( \mathbf{W}_{\text{RF}}^{(k)} \right)^H \mathbf{R}_{\text{YY}} \mathbf{W}_{\text{RF}}^{(k)} \right)^{-1} \left( \mathbf{W}_{\text{RF}}^{(k)} \right)^H \mathbf{R}_{\text{YY}} \mathbf{W}_{\text{MMSE}}$ 
     $\mathbf{W}_{\text{res}}^{(k)} = \frac{\mathbf{W}_{\text{MMSE}} - \mathbf{W}_{\text{RF}}^{(k)} \mathbf{W}_{\text{BB}}^{(k)}}{\left\| \mathbf{W}_{\text{MMSE}} - \mathbf{W}_{\text{RF}}^{(k)} \mathbf{W}_{\text{BB}}^{(k)} \right\|_{\text{F}}}$ 
end
 $\mathbf{W}_{\text{BB}} = \mathbf{W}_{\text{BB}}^*$ ,  $\mathbf{W}_{\text{RF}} = \mathbf{W}_{\text{RF}}^*$ 

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2.8.2. Deep Learning Framework for Optimal Beamforming

This section discusses how deep learning can be used to realize intelligent hybrid precoding in mmWave MISO systems. Deep neural networks' superior learning abilities allow the system to take advantage of the sparsity features of the frequency range 2 (FR2: 24GHz to 100GHz) band. Although there are numerous benefits of using deep learning, the ability to use spatial features for hybrid precoding without having to go into the system intricacies is unquestionable. The difficulties in using DL to solve equation 2.10 with uncertain CSI were summarised.

We modelled the proposed DNN framework as shown in figure 2.7, which is capable of handling nonlinearity. Here we considered a 5 layer regular DNNs, however, other network structures, and DL strategies such as CNNs, RNNs, etc., can be indeed used in this framework.

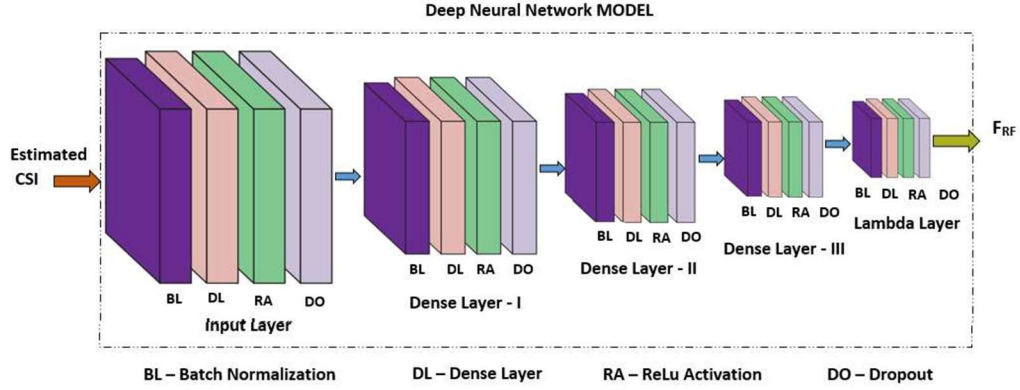


Figure 2.7 DNN Beamforming Model

This DNN model is installed at the base station, as it is required to utilize as few RF chains as possible, depending on the channel conditions the DNN may suffer from a bottlenecking effect in the flow of information through the layers of the neural network. However the model is trained for all possible samples of uncertain channel which makes the model to behave robust even in imperfect channel conditions.

Figure 2.8 describes the experimental flow in developing the DL mechanism adopted for mmWave MISO hybrid precoding and the training policy and testing model. The possible samples for perfect and imperfect CSI are generated as data set, which is fed as input to DNN model. The model is then trained for the given data set. After the training, DNN model is deployed online for practical channel conditions for testing. The beamforming weights produced after testing are considered to compute the spectral efficiency. The direct method of replacing the beamformer with a deep neural network and training it in the real time downlink encounter compatibility issues as the analog beamformer output is analogous [96], [97]. Here we propose a DL design that directly outputs F_{RF} to solve equation 2.10. We also design the DL model to generate an optimized analog beamformer vector F_{RF} based on the input of the channel estimate and SNR estimate.

We assume the base station has a channel estimator and use the estimated CSI H as the neural network's input. The DNN is anticipated to learn how to optimise the beamformer with uncertain CSI via the training process, as the input does not need to be ideal CSI. The SNR value associated with each sample is also input to the neural network, which improves generalisation ability for diverse

SNRs.

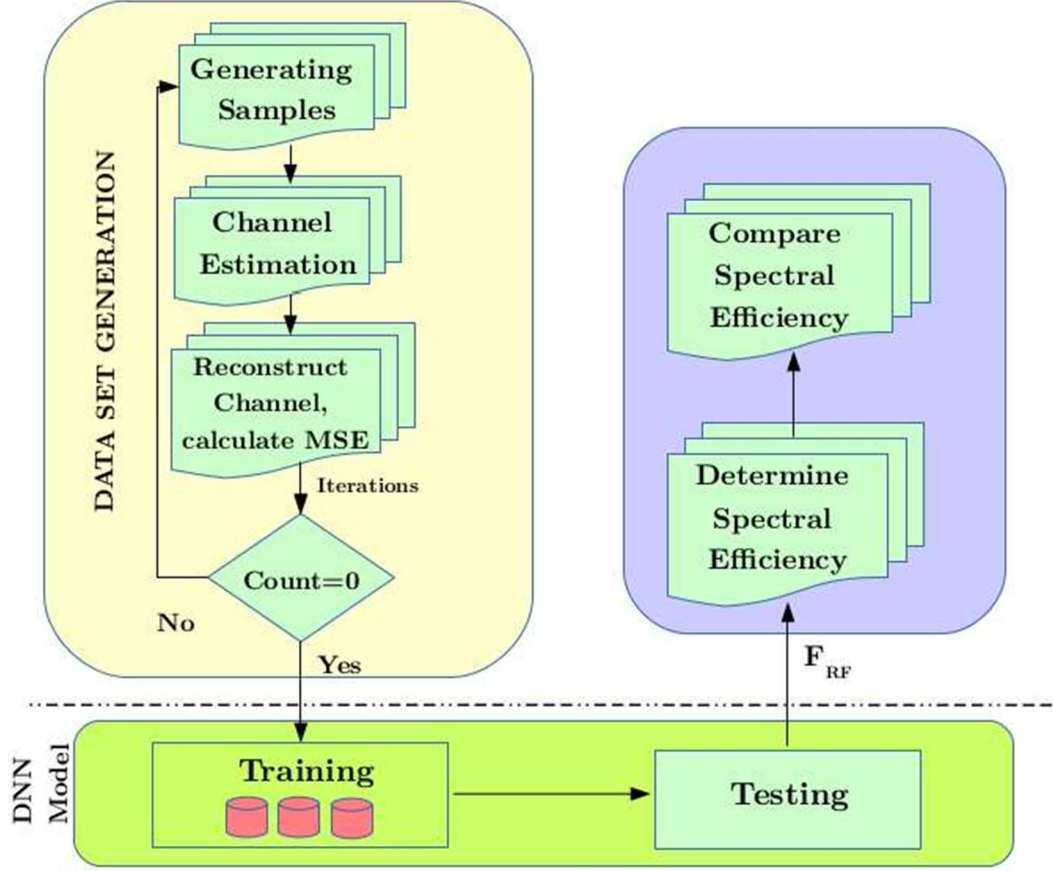


Figure 2.8 Flow of DL beamforming mechanism

The F_{RF} should be the complex vector to satisfy the constant modulus constraint, but most DL frameworks do not support the complex outputs (F_{RF}). Handling the constant modulus limitation: we propose adding a new lambda layer to the end of DNN to ensure that the output analog beamformer meets the constant modulus constraint. Furthermore, we present a self-defined loss function for key performance metrics, particularly spectral efficiency that does not require explicit labelling. The MSE between the exact and generated signals is the loss function as given below in equation 2.12.

$$Loss = -\frac{1}{N} \sum_{n=1}^N \log_2(1 + \frac{\gamma}{N_t} \|\mathbf{H}_k^H \mathbf{F}_{RF}\|^2) \quad (2.12)$$

The activation functions sigmoid and rectified linear unit (ReLU) are popularly used in nonlinear operations. Considering the argument variable as a , these activation functions are defined as $ReLU(a) = \max(0, a)$ and $Sigmoid(a) = \frac{1}{1+e^{-a}}$. Considering v as the input data, the mapping

operation can be presented as

$$z = f(v, w) = f^{(n-1)}(f^{(n-2)}(\dots f^1(v))) \quad (2.13)$$

Where n and w represent the number of layers in the neural networks and the weights of the neural network, respectively.

We can't utilise the traditional method to replace the analog beamformer with a neural network and train it directly online in the communication channel between the BS and the user since the analog beamformer has a special architecture that includes analog phase shifters. As a result, the three specific aspects are taken into account in our suggested model: The channel estimator, which forecasts the channel information, feeds the DNN. Second, the lambda layer was added to the DNN design to satisfy the analog beamformer's constant modulus restriction. Third, unlike the typical supervised learning approaches in [89], [92], and similar to the unsupervised learning scheme in [107], the loss function in our design does not require labels. The loss function in equation 2.12 is used to train the DNN model, which helps to achieve the goal represented by equation 2.10. With the foregoing in mind, the proposed DNN Beamforming framework is depicted in figure 2.7 in two stages: training and testing. The model is trained offline, with the first channel samples generated by simulation according to the MISO system model. At the BS, mmWave channel estimator is used to obtain imperfect CSI. The DNN model constructs the optimum analog beamforming vector F_{RF} by reducing the loss function defined in equation 2.12 using the estimated channel information H as input. The model is trained to figure out how to get close to the optimal spectral efficiency while maintaining perfect CSI and being resistant to channel estimation mistakes. The same channel estimator is utilised at the BS during the online testing stage. In the training stage, all parameters of the DNN model were trained for perfect CSI; in online testing mode, the model accepts imperfect CSI as input and directly outputs analog beamforming weights.

2.9. Results

The performance comparison of the proposed ESOMP, SOMP, and DBF schemes are presented in table 2.4. System capacity in bits/sec/Hz is evaluated for these three schemes. The SOMP scheme is based on compressive sensing that examines the sparsity

characteristics due to scatterers in the channel but permits relatively more errors. The proposed ESOMP method controls the error rate and hence renders better

capacity than SOMP and is near to the ideal case of DBF. The greater spectral efficiency in ESOMP is achieved by virtue of the concept of angular grids adopted in ESOMP. The angular grids are not considered in SOMP, and the SOMP precoder design assumes an ideal precoder that do not capture the mmWave characteristics, further the precoder is formulated as linear combination of columns of array transmit vectors. On the other hand, the robust performance of DNN based beamforming framework is accomplished by setting the model with the parameter values mentioned in table 2.5. A detailed discussion on these results is provided in chapter-5.

Table.2.4 Performance comparison of hybrid beamforming schemes

System Capacity in bits/Sec/Hz				
Data Streams, RF Chains	Antenna Type	Beamforming Techniques		
		DBF	SOMP HBF	Proposed ESOMP HBF
Ns=4, $N_t^{R\neq} 4$	ULA ($N_t = 64, N_r = 16$)	18	5.2	10.4
	UPA (Tx = 8×8 , Rx = 4×4)	27	16	19
Ns=6, $N_t^{R\neq} 6$	ULA ($N_t = 64, N_r = 16$)	18	8.5	14.2
	UPA (Tx = 8×8 , Rx = 4×4)	32	23	28

Table.2.5 Implementation details of DNN beamforming model

Layer Name	Batch Normalization				Output Dimension		Activation Function	Number of Parameters	
	Gamma	Beta	Moving Mean	Moving Variance	Bias	Kernel		Batch Normalization	Dense
Input Layer	129	129	129	129	4096	129x4096	----- -	532	516480

Dense Layer I	4096	4096	4096	4096	4096	4096x2048	ReLU	16384	8390656
Dense Layer II	2048	2048	2048	2048	2048	2048x1024	ReLU	8192	2098176
Dense Layer III	1024	1024	1024	1024	1024	1024x256	----- -	4096	262400
λ Layer	256	256	256	256	256	64	----- -	1024	16448

2.10. Summary

For the downlink case, this chapter presents a statistical hybrid beamforming approach that generates the best precoding and combining weights for mmWave MIMO communications. The suggested approach provides spectral efficiency comparable to a fully digital beamforming technique while decreasing hardware complexity, cost, and power consumption.

The DL based beamforming framework for mmWave MISO system presented in this chapter manages the challenges on the hardware constraints leveraging the spatial characteristics of mmWave channels. The DL scheme is proven to be more robust in imperfect channel conditions and provide superior performance in terms of spectral efficiency when compared to conventional beamforming method.

Next chapter brings power reduction models for massive MIMO systems. A comprehensive study of power scaling methods for perfect and imperfect channel conditions is done. The factor by which the amount of transmitting power can be saved is determined under both channel conditions. The quantity of power that can be save is relative to the number of antennas used in the system.